AI-Powered Virtual Try-On System for Raritone

Final Internship Documentation

Executive Summary

This project focused on researching and designing an AI-powered Virtual Try-On (VTO) system for fashion e-commerce. The goal was to improve online shopping by allowing users to virtually try garments before purchasing.

Project Objectives

Study Virtual Try-On technologies, analyze AI/ML models, evaluate datasets, explore recommendation systems, and provide implementation recommendations for Raritone.

Research Completed

Research covered Human Pose Estimation, Human Parsing, Image Segmentation, Cloth Warping, Garment Fitting, Recommendation Systems, Augmented Reality (AR), Virtual Reality (VR), and 3D visualization technologies.

Human Pose Estimation

MediaPipe, OpenPose, and Detectron2 were studied. MediaPipe was identified as the best choice for real-time applications because of its speed and lightweight architecture.

Image Segmentation

U2Net, Mask R-CNN, and Detectron2 were analyzed. Accurate segmentation is critical for separating users, garments, and backgrounds before virtual try-on processing.

Cloth Warping and Garment Fitting

TPS, GMM, Flow-Based Warping, and DensePose-based fitting techniques were compared. DensePose improves garment alignment by understanding body surface information.

Datasets Collected

DeepFashion, DeepFashion2, VITON, VITON-HD, MPV, and ModaNet datasets were reviewed for training and evaluation purposes.

2D Virtual Try-On Findings

Advantages include lower computational cost, faster implementation, and suitability for web applications. Limitations include reduced realism and restricted viewing angles.

3D Virtual Try-On Findings

3D systems provide realistic garment fitting, body-aware simulation, and immersive shopping experiences using AR/VR technologies.

AI Recommendation System

A hybrid recommendation system combining Content-Based Filtering, Collaborative Filtering, and Deep Learning was proposed to improve personalization and product discovery.

Challenges and Solutions

Challenges included body shape variations, cloth alignment issues, dataset limitations, and real-time performance requirements. Solutions involved DensePose, TPS, GMM, U2Net, and MediaPipe.

Complete AI Workflow

User Upload → Background Removal → Human Parsing → Pose Estimation → Garment Segmentation → Cloth Warping → Virtual Try-On Generation → Recommendation Engine → Final Output.

Model Comparison

MediaPipe: Fast; OpenPose: Accurate; Detectron2: Powerful segmentation; U2Net: Excellent background removal; DensePose: Detailed body mapping.

Future Improvements

Real-time webcam try-on, mobile deployment, AI fashion assistant, 3D avatar generation, AR smart mirrors, and cloud-based scalability.

Recommendations for Raritone

Begin with a 2D prototype using MediaPipe and U2Net, then progressively introduce 3D try-on, AR experiences, and advanced recommendation engines.

Conclusion

The project demonstrates that AI-powered Virtual Try-On systems can significantly improve customer engagement, conversion rates, and personalization in online fashion retail.